

PropVest

Product Requirements Document (PRD)

*The complete blueprint for building PropVest
Every screen, every feature, every API endpoint*

CONFIDENTIAL — Internal Document

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PropVest — Product Requirements Document (PRD)

The complete blueprint for building PropVest. Every screen, every feature, every API endpoint. Organized by platform and tagged by build phase.

1. PRODUCT OVERVIEW

1.1 Vision

Make real estate investment accessible to every Nigerian by letting trusted developers fractionally sell their properties to their own followers.

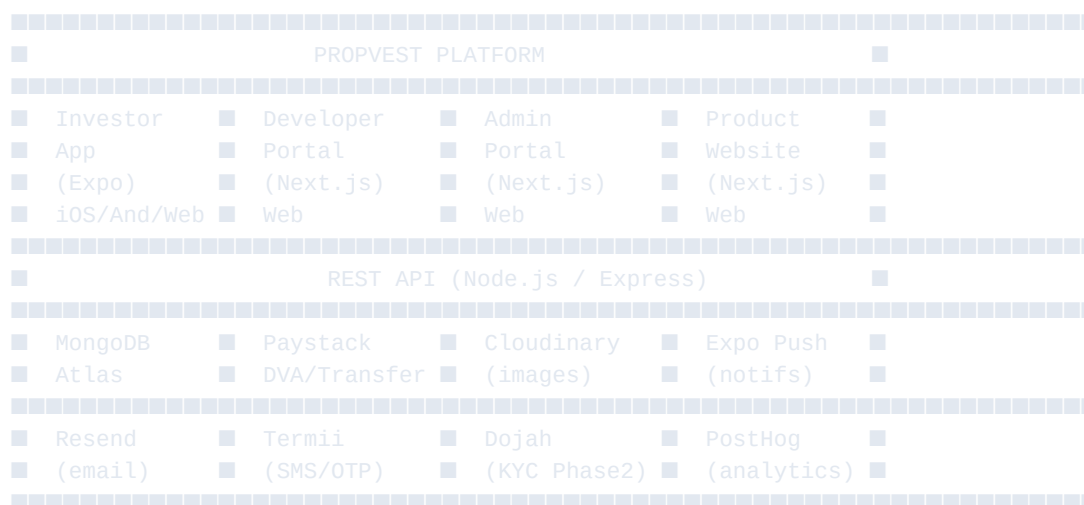
1.2 Mission

Build the invisible infrastructure that lets property developers raise capital from their Instagram followers at N100,000 per unit, while handling all the boring stuff — payments, KYC, payouts, legal, and marketplace.

1.3 Target Users

User Type	Who They Are	What They Want
Investors	Nigerian workers, diaspora, anyone with N50K-N5M to invest	Own a piece of verified property, earn rent or profit, grow wealth
Developer Partners	Established property developers with social media following (Dan Mama, Saleem Goje, etc.)	Raise capital faster from their audience without bank loans
Platform Admins	PropVest staff (CEO, COO, Operations, Finance, Legal, Support)	Manage the platform, investors, properties, payouts, compliance

1.4 Platform Architecture



1.5 Tech Stack

Layer	Technology
Investor App	React Native + Expo (iOS, Android, Web)
Developer Portal	Next.js 15 + TypeScript + Tailwind CSS
Admin Portal	Next.js 15 + TypeScript + Tailwind CSS
Product Website	Next.js 15 + TypeScript + Tailwind CSS
Backend API	Node.js + Express (or Next.js API routes)
Database	MongoDB Atlas
Auth (web)	NextAuth.js with JWT
Auth (mobile)	Custom JWT + secure storage
Payments In	Paystack DVA (Dedicated Virtual Accounts)
Payment Split	Paystack Sub-accounts (auto-split: developer gets 97%, PropVest gets 3%)
Payments Out	Paystack Transfers (Bulk)
KYC Tier 1	Paystack Resolve Account (free)
KYC Tier 2	Paystack BVN Resolve (N10/call)
KYC Tier 3	Dojah NIN + facial (Phase 2)

Layer	Technology
Email	Resend
SMS / OTP	Termii
Push Notifications	Expo Push Notifications
Image Storage	Cloudinary
Document Storage	Cloudinary (signed URLs for private docs)
Analytics	PostHog (self-hosted or cloud)
Hosting (web)	Vercel
Hosting (mobile)	EAS Build + EAS Submit

1.6 Design System

Full visual design specification is in **DESIGN_SYSTEM.md** — covers color palette, typography (Inter + JetBrains Mono), spacing system, component library, animations, dark mode, screen layouts with ASCII mockups, accessibility standards, and platform-specific guidelines. Every component, every color, every animation is documented.

2. INVESTOR APP (React Native + Expo)

iOS + Android + Web. This is what investors use to browse, invest, track, and earn.

2.1 Authentication

Screen	Features	Phase
Welcome / Onboarding	3-slide intro (what is PropVest, how it works, get started). Skip button.	MVP

Screen	Features	Phase
Sign Up	Full name, email, phone number, password, confirm password. Terms checkbox.	MVP
OTP Verification	6-digit OTP sent via SMS (Termii). Auto-read on Android. Resend button with 60s cooldown.	MVP
Login	Email/phone + password. "Forgot password" link. Biometric login toggle (after first login).	MVP
Forgot Password	Enter email → receive reset link → set new password.	MVP
Biometric Login	Fingerprint / Face ID. Enable in settings after first successful login.	Phase 1

2.2 KYC Onboarding (Tiered)

Tier	What User Provides	API Used	Cost	Limit	Phase
Tier 1	Bank account number + bank name	Paystack Resolve Account	Free	Invest up to N500,000	MVP
Tier 2	BVN (11 digits)	Paystack BVN Resolve	N10	Invest up to N5,000,000	Phase 1
Tier 3	NIN + selfie photo	Dojah NIN + Liveness	~N150	Unlimited	Phase 2

KYC Flow:

After signup → "Verify your identity to start investing"
 → Enter bank name (dropdown) + account number
 → Backend calls Paystack Resolve → returns name
 → Match name with signup name
 → If match → Tier 1 verified → can browse and invest
 → If mismatch → show error, ask to retry or contact support

When investor tries to exceed N500,000 total:
 → "Upgrade your verification to invest more"
 → Enter BVN → backend verifies → Tier 2 unlocked

When investor tries to exceed N5,000,000:

→ "Complete full verification"

→ Enter NIN + take selfie → Dojah verifies → Tier 3 unlocked

2.3 Home Dashboard

Element	Description	Phase
Portfolio summary card	Total invested, current value, total returns earned, unrealized gains (%). Tap to go to portfolio.	MVP
Recent payouts	Last 3 payouts with amount and date. "See all" link.	MVP
Trending properties	Top 3 properties by investor count or % funded. Horizontal scroll cards.	MVP
AI recommended	"Recommended for you" section — 3 properties based on behavior.	Phase 2
New from followed developers	Properties from developers you follow. Only shows if you follow someone.	Phase 1
Quick actions	Buttons: "Invest", "My Portfolio", "Marketplace", "Payouts"	MVP
Weekly summary banner	"Your portfolio is up X% this week" — dismissable.	Phase 2
Announcements	Admin-pushed announcements (maintenance, new features, etc.)	Phase 1

2.4 Property Browsing

Element	Description	Phase
Tabs	"All", "Rental", "Build & Sell", "Land"	MVP
Search	Search by property name, location, developer name	MVP

Element	Description	Phase
Filters	Location, price range (min unit price), property type, developer, % funded, sort by (newest, most popular, highest yield)	MVP
Property card	Image, name, developer name, location, unit price, % funded progress bar, property type badge (Rental / Build & Sell / Land), estimated return	MVP
Sold out badge	Grey overlay "Fully Funded" on properties with no units left	MVP
Wishlist icon	Heart icon on each card — save to wishlist	Phase 1

2.5 Property Detail

Section	Content	Phase
Image gallery	Swipeable property photos. Tap to fullscreen.	MVP
Header	Property name, developer name (tappable → developer profile), location, property type badge	MVP
Funding progress	Progress bar, "X of Y units sold", "Z% funded", units remaining	MVP
Key numbers	Unit price, total property value, estimated annual return (%), payout frequency (quarterly / on sale)	MVP
About	Property description written by developer. Construction details for build & sell.	MVP
Financials tab	For rental: annual rent, quarterly payout per unit, rent yield %. For build & sell: build cost, estimated sale price, estimated profit %, timeline. For land: price per sqm, total area, milestone payout schedule.	MVP

Section	Content	Phase
Documents tab	C of O, Governor's Consent, building plans, inspection report. View / download.	MVP
Developer updates tab	Timeline of updates posted by developer (construction photos, milestones).	Phase 1
Investor count	"X investors have invested in this property" (social proof)	MVP
Invest CTA	Sticky bottom button: "Invest Now" → goes to investment flow	MVP
Share button	Share property via WhatsApp, copy link, Instagram story	Phase 1
Wishlist button	Save / unsave from wishlist	Phase 1

2.6 Investment Flow

Step 1: SELECT UNITS

- Property summary (name, unit price, remaining units)
- Unit quantity input (+ / - buttons, or type number)
- Total amount auto-calculated
- "Continue" button (disabled if quantity is 0 or exceeds KYC limit)

Step 2: REVIEW & CONFIRM

- Summary: property name, X units at N100,000 each = N[total]
- Platform fee: 2% = N[amount]
- Total to pay: N[total + fee]
- Checkbox: "I understand this is an investment with risks"
- "Confirm & Pay" button

Step 3: PAYMENT

- Show Paystack DVA bank account details:
 - Bank: Wema Bank (or Titan)
 - Account Number: 7821093456
 - Account Name: PropVest/[Property Name]
 - Amount: N[total]
 - Reference: INV-XXXXXX
- "I have made the transfer" button
- Or: countdown timer (DVA account expires in 30 min)
- Waiting state: "We're confirming your payment..."

PAYMENT ROUTING (behind the scenes):

- Paystack Sub-account is pre-configured per property
- On payment received, Paystack auto-splits:
 - 97% (property share) settles to developer's bank account
 - 3% (platform fee) settles to PropVest's bank account
- Developer sees money arrive in their own bank account directly

- PropVest never holds developer's money

Step 4: CONFIRMATION

- Success screen with confetti/animation
- "You now own X units in [property name]!"
- Summary: units owned, expected returns
- Buttons: "View Portfolio", "Share with friends", "Invest in more"

Phase: MVP (all steps)

2.7 Portfolio

Tab	Content	Phase
Holdings	List of all properties you own units in. Per property: name, units owned, current value, return %, property type badge. Tap → property detail.	MVP
Performance	Total portfolio value over time (line chart). Total returns earned (bar chart by quarter).	Phase 1
Activity	Timeline of all investment activity: purchases, payouts received, units sold, units bought.	MVP

Per holding detail (tap on a holding):

- Units owned, purchase price, current value, gain/loss
- Payout history for this property
- "Sell Units" button → goes to marketplace listing flow
- Developer updates for this property

2.8 Payouts

Element	Description	Phase
Upcoming payouts	Next expected payout: property name, estimated amount, estimated date	MVP
Payout history	List of all payouts received: date, property, amount, status (sent / pending / failed)	MVP

Element	Description	Phase
Total earned	Lifetime total payouts received	MVP
Bank accounts	List of linked bank accounts. Add / remove. One marked as "primary" for payouts.	MVP
Payout breakdown	Per payout: which property, how many units, rent amount, management fee deducted, your share	Phase 1

2.9 Secondary Market (P2P)

Screen	Content	Phase
Browse listings	All units for sale. Filter by property, price range, developer. Show: property name, seller's asking price, original price, gain/loss %, units available.	MVP
Buy flow	Tap "Buy" → review (units, price, trading fee) → pay via DVA → units transferred. Same payment flow as investment.	MVP
Sell flow	From portfolio → "Sell Units" → set price per unit, select how many units → list on marketplace. Can cancel listing anytime.	MVP
My listings	View your active sell listings. Edit price, cancel listing.	MVP
Price alerts	Get notified when units in a property you own are listed below your purchase price	Phase 2
Trade history	All completed P2P trades (bought and sold)	MVP

2.10 Developer Profiles

Element	Description	Phase
Developer page	Developer name, logo, bio/description, follower count, properties listed, total raised, investor count	Phase 1
Follow button	Follow / unfollow. Get notified of new listings.	Phase 1
Properties tab	All properties by this developer (active, fully funded, completed)	Phase 1
Track record	Completed properties, average returns delivered, payout history	Phase 2
Updates tab	All updates posted by this developer across all properties	Phase 1

2.11 Notifications

Notification center (bell icon on home screen):

- List of all notifications, newest first
- Read/unread indicator (blue dot)
- Tap to navigate to relevant screen
- "Mark all as read"
- Filter: All, Payouts, Properties, Marketplace, System

Notification types (detailed in plan section 2a):

Category	Notifications	Delivery	Phase
Transactional	Payment confirmed, payout sent, P2P completed, KYC status	Push + In-app + Email	MVP
Property Updates	New listing from followed dev, funding progress, value update, construction milestone, rent collected, property sold	Push + In-app	Phase 1

Category	Notifications	Delivery	Phase
Engagement	New property in your area, appreciation stats, inactivity nudge, portfolio summary, urgency ("X units left"), milestone celebration	Push	Phase 2
AI Recommendations	Similar properties, "investors like you", pattern matching, track record alerts, marketplace price alerts	Push + In-app	Phase 2

Notification preferences (in settings):

- Toggle each category on/off
- Toggle push / email / SMS separately
- Quiet hours (no push between 10pm - 7am)

Phase: MVP (transactional only), Phase 1 (property updates), Phase 2 (engagement + AI)

2.12 Profile & Settings

Section	Content	Phase
Personal info	Name, email, phone (read-only after KYC)	MVP
KYC status	Current tier, upgrade prompt, verification history	MVP
Bank accounts	Add / remove / set primary. Verified via Paystack.	MVP
Security	Change password, enable biometric login, active sessions	MVP
Notification preferences	Toggle notifications by category	Phase 1
Referral	Your referral code, share link, referral history, rewards earned	Phase 2
Document vault	Investment certificates, investment receipts, tax documents	Phase 2

Section	Content	Phase
Investment calculator	Input amount → see projected returns (rental yield, build & sell profit, land appreciation)	Phase 1
Help & support	FAQ, in-app chat, email support, WhatsApp link	MVP
Legal	Terms of Service, Privacy Policy, Investment Risks	MVP
App settings	Dark mode toggle, language (English), currency (NGN)	Phase 1
Delete account	Request account deletion (NDPA requirement)	MVP

2.13 Additional Features

Feature	Description	Phase
Wishlist	Save properties to invest later. Accessible from profile.	Phase 1
Share property	Share via WhatsApp, copy link, Instagram story, Twitter	Phase 1
Investment calculator	Slider: amount → projected quarterly rent, annual return, 5-year projection	Phase 1
In-app support chat	Real-time chat with support team	Phase 2
Dark mode	System-default or manual toggle	Phase 1
Biometric login	Fingerprint / Face ID	Phase 1
Referral system	Invite friends → earn bonus on their first investment	Phase 2
Offline mode	View portfolio and holdings offline (cached data)	Phase 2

3. DEVELOPER PORTAL (Next.js)

Web application for developer partners. This is where Saleem Goje or Dan Mama manages their properties and investors.

3.1 Authentication

Feature	Description	Phase
Invite-only signup	Admin sends invite link → developer creates account	MVP
Login	Email + password	MVP
2FA	OTP via SMS on login (for security — they manage money)	Phase 1

3.2 Dashboard

Element	Description	Phase
Stats cards	Properties listed, total raised, active investors, total followers on platform, pending payouts	MVP
Revenue summary	Your total earnings (from retained units), rent received, management fee split	MVP
Properties overview	Quick list of all properties with funding status, units sold, next payout date	MVP
Recent activity	Latest investments, new followers, recent payouts	Phase 1
Notifications	Pending actions: rent due, payout approval needed, new investor messages	MVP

3.3 Property Management

Property list:

- All properties with status: Draft, Under Review, Active, Fully Funded, Completed
- Filter by status, type (rental, build & sell, land)

Per property detail page:

Section	Content	Phase
Overview	Property details, funding progress, investor count, unit price	MVP
Investors	List of all investors in this property: name, units, amount invested, date. Export to CSV.	MVP
Funding progress	Visual progress bar, units sold over time chart	MVP
Payouts	All payouts for this property: date, total distributed, your share, status	MVP
Documents	Uploaded documents (C of O, plans, photos). Add / replace.	MVP
Updates	Post updates to investors (text + photos). Shows on investor app.	Phase 1
Valuation	Current property value. Update value (with justification). Requires admin approval.	Phase 1
Sales reporting	For build & sell: report houses sold, upload proof, trigger profit distribution	MVP
Rent reporting	For rental: report rent collected, upload proof, trigger payout preparation	MVP

3.4 Add New Property

Step 1: BASIC INFO

- Property name
- Property type (Rental / Build & Sell / Land)
- Location (state, city, area)
- Description (rich text)
- Upload photos (min 3, max 20)

Step 2: FINANCIAL DETAILS

For Rental:

- Property value
- Annual rent
- Target number of units
- Unit price (auto-calculated or manual)
- Developer retained units (how many you keep)

For Build & Sell:

- Total build cost
- Estimated sale price per house
- Number of houses
- Target number of units
- Unit price
- Estimated timeline (months)
- Developer retained units

For Land:

- Total area (hectares / sqm)
- Price per sqm
- Minimum investment (sqm)
- Plot conversion threshold (e.g., 500 sqm)
- Milestone payout schedule (25%, 50%, 75%, 100% sold)

Step 3: DOCUMENTS

- Upload C of O / Governor's Consent
- Upload building plans / survey plans
- Upload any inspection reports
- Upload proof of ownership

Step 4: REVIEW & SUBMIT

- Review all entered information
- Submit for admin review
- Status: "Under Review"

Phase: MVP

3.5 Marketing Toolkit

Feature	Description	Phase
Branded link	Unique URL for each property: p ropvest.ng/invest/saleem-goje/g warinpa-duplex	MVP
QR code	Downloadable QR code linking to property page	Phase 1
Social media graphics	Auto-generated Instagram post/story templates with property photo, price, and CTA	Phase 2

Feature	Description	Phase
Embed widget	Code snippet to embed investment CTA on developer's own website	Phase 2
Share analytics	Track how many clicks from shared links, which social platform, conversion rate	Phase 2

3.6 Investor Analytics

Metric	Description	Phase
Total investors	Across all properties	MVP
Average investment size	Helps developer understand their audience	Phase 1
Investor demographics	Location breakdown (state), investment frequency	Phase 2
Follower growth	Chart of followers over time on PropVest	Phase 1
Conversion rate	Followers → investors (how many followers actually invest)	Phase 2

4. ADMIN PORTAL (Next.js) — Role-Based Access

Web application for PropVest staff. Every feature is access-controlled based on role.

4.1 Role-Based Access Control (RBAC)

Feature / Page	CEO	COO	Ops Associate	Finance	Legal	DevRel	Support	Compliance
Dashboard	Full	Full	Limited	Financial	Legal	DevRel	Support	Compliance
Properties	CRUD	CRUD	Read	Read	Read+Docs	CRUD	Read	Read
Approve listing	Yes	Yes	No	No	Yes (docs)	Recommend	No	No
Investors	Full	Full	Read+KYC	Read	Read	Read	Read	Full
KYC Queue	Full	Full	Process	No	No	No	View	Audit
Payouts	Full	Approve	View	Full	No	No	No	Audit
Trigger payout	Yes	Yes	No	Prepare	No	No	No	No
P2P Trades	Full	Full	Monitor	Read	No	No	Disputes	Monitor
Developer Partners	Full	Full	Read	Read	Contracts	Full	Read	Read
Financial Reports	Full	Full	No	Full	No	No	No	Read
Compliance	Full	Read	No	No	Full	No	No	Full
Staff Management	Full	No	No	No	No	No	No	No
System Settings	Full	Limited	No	No	No	No	No	No
Audit Log	Full	Full	No	No	Full	No	No	Full
Announcements	Full	Full	No	No	No	No	Draft	No

4.2 Admin Dashboard

Element	Description	Phase
Key metrics	Total investors, total properties, total assets managed, platform revenue (MTD, YTD), total payouts distributed	MVP
Pending actions	KYC approvals pending, properties awaiting review, payouts to process, support tickets open	MVP
Revenue chart	Platform fee + management fee + trading fee over time	Phase 1
Recent transactions	Last 20 transactions across all properties	MVP
Alerts	Failed payouts, flagged transactions, developer rent overdue, KYC issues	MVP
Quick actions	"Review KYC", "Process Payouts", "View Properties"	MVP

4.3 Properties Management

Feature	Description	Phase
List all properties	Table: name, developer, type, status, % funded, investors, total raised. Search + filters.	MVP
Property detail	All info: financials, investors, payouts, documents, updates, payment routing details	MVP
Approve/reject listing	Review submitted property. Check documents. Approve or reject with reason.	MVP
Payment routing setup	Create Paystack sub-account for property, configure split ratio, link developer bank account, generate DVA	MVP
Edit property	Update details, change status (active → paused → completed)	MVP

Feature	Description	Phase
Property health	Flag: rent overdue, construction delayed, investor complaints	Phase 1

4.4 Investor Management

Feature	Description	Phase
List all investors	Table: name, email, phone, KYC tier, total invested, properties count, join date. Search + filters.	MVP
Investor detail	Profile, KYC info, all investments, payout history, support tickets, activity log	MVP
Freeze account	Temporarily freeze investor account (suspected fraud, compliance issue)	MVP
Flag account	Flag for review (doesn't freeze, just marks for attention)	MVP
Export	Export investor list to CSV (for compliance reporting)	Phase 1
Send notification	Send push/email to specific investor	Phase 1

4.5 KYC Queue

Feature	Description	Phase
Pending queue	List of investors awaiting KYC review (mainly Tier 2-3 or flagged Tier 1)	MVP
Review screen	Show investor's submitted info vs API-returned info. Match/mismatch highlights.	MVP
Approve / Reject / Request more info	Actions with reason field	MVP

Feature	Description	Phase
Auto-approve	Tier 1 (bank account match) auto-approves. Only mismatches go to queue.	MVP
KYC stats	Approved today, rejected today, pending, average processing time	Phase 1

4.6 Payout Management

Feature	Description	Phase
Upcoming payouts	Calendar/list of upcoming quarterly payouts per property	MVP
Prepare payout	Select property → system calculates each investor's share → review breakdown	MVP
Approve payout	CEO or COO approves → Finance triggers Paystack Bulk Transfer	MVP
Monitor payout	Real-time status of all transfers (sent, pending, failed)	MVP
Retry failed	Retry individual failed transfers (wrong bank details, etc.)	MVP
Payout history	All past payouts: date, property, total distributed, fees, status	MVP
Reconciliation	Compare: rent received from developer vs total distributed to investors vs fees collected. Flag discrepancies.	Phase 1
Payout reports	Downloadable PDF/CSV per payout cycle	Phase 1

4.7 P2P Trade Monitoring

Feature	Description	Phase
Active listings	All units currently listed for sale	MVP

Feature	Description	Phase
Completed trades	All completed P2P transactions with fee collected	MVP
Flagged trades	Auto-flagged: same buyer/seller, unusual price, high frequency	Phase 1
Dispute queue	Investor-reported disputes. Investigate, resolve, or escalate.	MVP
Trading stats	Volume, average markup, most traded properties	Phase 1

4.8 Developer Partner Management

Feature	Description	Phase
List all developers	Table: name, properties, total raised, investors, follower count, status	MVP
Developer detail	Profile, all properties, payout history, documents, notes	MVP
Onboard new developer	Create account, send invite, upload partnership agreement	MVP
Performance dashboard	Per developer: funding speed, rent collection timeliness, investor satisfaction	Phase 1
Rent tracking	Track expected rent dates. Auto-flag if rent is X days overdue.	Phase 1
Communication log	Notes from interactions with developer (calls, meetings, issues)	Phase 1

4.9 Financial Reports

Report	Description	Phase
Revenue dashboard	Platform fees, management fees, trading fees — by month, by property	MVP

Report	Description	Phase
Expense tracking	Paystack fees, KYC costs, other operational costs	Phase 1
Per-property P&L	Revenue vs costs per property	Phase 1
Per-property financial report	Total collected via Paystack, developer's share, platform fees earned, payouts distributed, current status	MVP
Tax report	VAT collected, WHT applicable, CIT calculation helper	Phase 2
Investor report	Total invested, total returned, platform growth metrics	Phase 1

4.10 Compliance

Feature	Description	Phase
Transaction monitoring	Flag large transactions (above threshold), unusual patterns	Phase 1
STR filing	Suspicious Transaction Report form (for NFIU submission)	Phase 2
KYC audit	Review KYC decisions, spot checks, compliance statistics	Phase 1
AML dashboard	Risk scores for investors based on transaction patterns	Phase 2
Regulatory documents	Store SEC filings, SCUML certificate, NDPA audit reports	Phase 1

4.11 Staff Management

Feature	Description	Phase
List staff	All admin users: name, email, role, last active, status	MVP
Add staff	Create new admin account with role assignment	MVP

Feature	Description	Phase
Edit role	Change staff member's role (changes their access immediately)	MVP
Deactivate	Disable staff account without deleting (preserves audit trail)	MVP
Activity log	Per staff member: what they accessed, what they changed	Phase 1

4.12 System Settings

Setting	Description	Phase
Platform fee %	Default platform fee for new properties (adjustable per property)	MVP
Management fee %	Default management fee (adjustable per property)	MVP
Trading fee %	P2P marketplace fee	MVP
KYC tier limits	Max investment per KYC tier	MVP
Payout schedule	Default payout frequency (quarterly)	MVP
Maintenance mode	Toggle platform into maintenance mode (shows message to all users)	MVP
Feature flags	Enable/disable features (marketplace, referrals, etc.)	Phase 1

4.13 Audit Log

Every action by every user is logged:

- Who (user ID, name, role)
- What (action: create, update, delete, approve, reject, login, etc.)
- When (timestamp)
- Where (which page/feature)
- Details (what changed, old value → new value)

- IP address

Searchable, filterable, exportable. Cannot be deleted or modified.

Phase: MVP

4.14 Announcements & Notifications

Feature	Description	Phase
Broadcast alert	Send push notification + in-app banner to ALL users instantly. For urgent things: payment gateway down, scheduled maintenance, security alerts. Simple text input + send.	MVP
Send announcement	Push notification to all investors or filtered group (by property, by location, etc.)	Phase 1
Schedule announcement	Send at a specific date/time	Phase 2
Email blast	Send email to all investors or filtered group	Phase 1
Notification templates	Pre-written templates for common notifications (payout sent, new property, etc.)	Phase 1

5. PRODUCT WEBSITE (Next.js)

Marketing website at propvest.ng. Explains the platform, attracts investors and developers.

Page	Content	Phase
Landing page	Hero with CTA ("Start Investing" / "List Your Property"), how it works (3 steps), key stats, featured properties, testimonials, developer section, FAQ preview, footer	MVP
For Investors	How investing works, property types explained, returns explained, risk disclosure, "Download the App" CTA	MVP
For Developers	How partnership works, benefits, onboarding process, "Apply to List" CTA	MVP
About	Company story, team, mission, values	Phase 1
FAQ	Expandable FAQ sections for investors and developers	MVP
Contact	Contact form, email, WhatsApp link, office address	MVP
Blog	SEO articles about real estate investment in Nigeria	Phase 2
Developer application	Form: company name, CAC number, website/Instagram, properties built, why they want to partner	MVP
Terms of Service	Legal terms	MVP
Privacy Policy	NDPA-compliant privacy policy	MVP
Investment Risks	Clear risk disclosure document	MVP

6. DATABASE SCHEMA (MongoDB)

6.1 Collections

users

```
{
  _id, email, phone, passwordHash,
  role: "investor" | "developer" | "admin",
  fullName, profileImage,
  kycTier: 1 | 2 | 3,
  kycStatus: "pending" | "verified" | "rejected",
  bankAccounts: [{ bankName, accountNumber, accountName, isPrimary }],
  adminRole: "ceo" | "coo" | "ops" | "finance" | "legal" | "devrel" | "support" | "compliance", // c
  isActive, isFrozen, isFlagged,
  referralCode, referredBy,
  pushToken, // Expo push token
  notificationPreferences: { payouts, properties, engagement, recommendations },
  biometricEnabled,
  createdAt, updatedAt, lastLoginAt
}
```

properties

```
{
  _id, developerId (ref: users),
  name, slug, description, location: { state, city, area, coordinates },
  type: "rental" | "build_sell" | "land",
  status: "draft" | "under_review" | "active" | "paused" | "fully_funded" | "completed",
  images: [url], documents: [{ name, url, type }],

  // Financials
  totalValue, // N90,000,000
  totalUnits, // 900
  unitPrice, // N100,000
  unitsSold,
  developerRetainedUnits,

  // Rental specific
  annualRent, rentYield, payoutFrequency: "quarterly",

  // Build & Sell specific
  buildCost, estimatedSalePrice, estimatedProfit, timelineMonths,
  housesTotal, housesSold,

  // Land specific
  totalAreaSqm, pricePerSqm, minInvestmentSqm, plotConversionThreshold,
  milestones: [{ percentSold, payoutPercent }],

  // Payment Routing (Paystack Split)
  paystackSubaccountCode, // Paystack sub-account for this property
  developerBankName,
  developerBankAccountNumber,
  developerBankAccountName,
  splitRatio: { developer: 97, platform: 3 }, // configurable per property

  // Meta
  platformFeePercent, managementFeePercent,
  currentValuation, // updated over time
  investorCount,
```

```

    createdAt, updatedAt, approvedAt, approvedBy
  }

```

investments

```

{
  _id, investorId (ref: users), propertyId (ref: properties),
  units, unitPrice, totalAmount,
  platformFee, amountAfterFee,
  transactionRef, // Paystack reference
  status: "pending" | "confirmed" | "failed",
  createdAt
}

```

transactions

```

{
  _id,
  type: "investment" | "payout" | "p2p_buy" | "p2p_sell" | "fee",
  userId (ref: users),
  propertyId (ref: properties),
  amount, fee, netAmount,
  paystackRef, paystackStatus,
  status: "pending" | "success" | "failed",
  metadata: {}, // flexible extra data
  createdAt
}

```

paymentRouting

```

{
  _id, propertyId (ref: properties), developerId (ref: users),
  paystackSubaccountCode, // Paystack sub-account code
  developerBankName,
  developerBankAccountNumber,
  developerBankAccountName,
  splitConfig: {
    developer: 97, // % to developer
    platform: 3 // % to PropVest (platform fee)
  },
  dvaAccountNumber, // Virtual account for this property
  dvaBank, // e.g. "Wema Bank"
  status: "active" | "paused",
  totalCollected, // running total of all payments received
  totalPlatformFee, // running total of PropVest's share
  createdAt
}

```

payouts

```

{
  _id, propertyId (ref: properties),
  quarter, // "2026-Q2"
  totalRentReceived,
}

```

```

managementFee, platformShare,
distributableAmount,
investorPayouts: [{
  investorId, units, share, amount,
  paystackTransferRef, status: "pending" | "sent" | "failed",
  bankName, accountNumber
}],
status: "calculating" | "pending_approval" | "approved" | "processing" | "completed" | "partial",
approvedBy, approvedAt,
triggeredBy, triggeredAt,
createdAt
}

```

marketplaceListings

```

{
  _id, sellerId (ref: users), propertyId (ref: properties),
  units, pricePerUnit,
  status: "active" | "sold" | "cancelled",
  buyerId (ref: users), // filled on sale
  tradingFee,
  createdAt, soldAt, cancelledAt
}

```

notifications

```

{
  _id, userId (ref: users),
  type: "transactional" | "property_update" | "engagement" | "recommendation" | "system",
  title, body,
  data: { propertyId, payoutId, etc }, // for deep linking
  channels: { push: bool, email: bool, sms: bool },
  pushSent, emailSent, smsSent,
  read: false,
  createdAt
}

```

follows

```

{
  _id, investorId (ref: users), developerId (ref: users),
  createdAt
}

```

propertyUpdates

```

{
  _id, propertyId (ref: properties), developerId (ref: users),
  title, body, images: [url],
  type: "general" | "construction" | "milestone" | "valuation",
  createdAt
}

```

kycRecords

```
{
  _id, userId (ref: users),
  tier, method: "bank_account" | "bvn" | "nin",
  submittedData: { bankName, accountNumber, bvn, nin },
  apiResponse: {}, // raw API response
  matchResult: "match" | "mismatch" | "partial",
  status: "pending" | "approved" | "rejected",
  reviewedBy, reviewedAt, reviewNotes,
  createdAt
}
```

referrals

```
{
  _id, referrerId (ref: users), referredId (ref: users),
  referralCode,
  status: "signed_up" | "invested" | "bonus_paid",
  bonusAmount, bonusPaidAt,
  createdAt
}
```

supportTickets

```
{
  _id, userId (ref: users),
  subject, status: "open" | "in_progress" | "resolved" | "closed",
  assignedTo (ref: users), // admin user
  messages: [{ senderId, body, createdAt }],
  priority: "low" | "medium" | "high",
  createdAt, resolvedAt
}
```

auditLogs

```
{
  _id, userId (ref: users),
  action, // "approve_kyc", "trigger_payout", "freeze_account", etc.
  resource, // "investor", "property", "payout", etc.
  resourceId,
  details: {}, // old value, new value
  ipAddress, userAgent,
  createdAt
}
```

wishlists

```
{
  _id, userId (ref: users), propertyId (ref: properties),
  createdAt
}
```

6.2 Indexes

Collection	Index	Purpose
users	email (unique)	Login lookup
users	phone (unique)	OTP and search
users	referralCode (unique)	Referral system
properties	slug (unique)	URL routing
properties	developerId + status	Developer portal queries
properties	status + type	Browse/filter
investments	investorId + propertyId	Portfolio queries
investments	propertyId	Per-property investor list
transactions	userId	Transaction history
transactions	paystackRef (unique)	Webhook matching
payouts	propertyId + quarter	Payout lookup
marketplaceListings	propertyId + status	Browse marketplace
follows	investorId	"Following" list
follows	developerId	Follower count
notifications	userId + read	Notification center
auditLogs	userId + createdAt	Admin activity
auditLogs	resource + resourceId	Resource history
paymentRouting	propertyId (unique)	Property payment config
paymentRouting	paystackSubaccountCode (unique)	Webhook matching

7. API ARCHITECTURE

7.1 Authentication Endpoints

POST	/api/auth/signup	Sign up with email, phone, password
POST	/api/auth/verify-otp	Verify phone OTP
POST	/api/auth/login	Login (returns JWT)
POST	/api/auth/refresh	Refresh JWT token
POST	/api/auth/forgot-password	Send reset link
POST	/api/auth/reset-password	Reset password with token
POST	/api/auth/biometric	Biometric login (validate token)

7.2 KYC Endpoints

POST	/api/kyc/verify-bank	Tier 1: bank account resolve
POST	/api/kyc/verify-bvn	Tier 2: BVN verification
POST	/api/kyc/verify-nin	Tier 3: NIN + selfie
GET	/api/kyc/status	Current KYC status

7.3 Property Endpoints

GET	/api/properties	List all (with filters, pagination)
GET	/api/properties/:id	Property detail
GET	/api/properties/:id/investors	Investors in property (admin/developer)
GET	/api/properties/:id/updates	Developer updates
GET	/api/properties/:id/documents	Property documents
POST	/api/properties	Create new property (developer)
PUT	/api/properties/:id	Update property (developer/admin)
POST	/api/properties/:id/approve	Approve listing (admin)
POST	/api/properties/:id/reject	Reject listing (admin)
POST	/api/properties/:id/updates	Post update (developer)
PUT	/api/properties/:id/valuation	Update valuation (developer, needs admin approval)

7.4 Investment Endpoints

POST	/api/investments	Create investment (initiate payment)
GET	/api/investments/my	My investments (portfolio)
GET	/api/investments/:id	Investment detail
POST	/api/webhooks/paystack	Paystack webhook handler

7.5 Payout Endpoints

GET	/api/payouts	List payouts (admin)
GET	/api/payouts/my	My payout history (investor)
POST	/api/payouts/calculate/:propertyId	Calculate payout (admin)
POST	/api/payouts/:id/approve	Approve payout (admin)

POST	/api/payouts/:id/trigger	Trigger Paystack transfers (admin)
POST	/api/payouts/:id/retry/:investorId	Retry failed transfer (admin)

7.6 Marketplace Endpoints

GET	/api/marketplace	Browse listings
POST	/api/marketplace	Create listing (sell units)
PUT	/api/marketplace/:id	Update listing (change price)
DELETE	/api/marketplace/:id	Cancel listing
POST	/api/marketplace/:id/buy	Buy listed units (initiate payment)
GET	/api/marketplace/my-listings	My active listings
GET	/api/marketplace/my-trades	My trade history

7.7 Developer Endpoints

GET	/api/developers/:id	Developer profile (public)
GET	/api/developers/:id/properties	Developer's properties
POST	/api/developers/:id/follow	Follow developer
DELETE	/api/developers/:id/follow	Unfollow developer
GET	/api/developers/following	My followed developers

7.8 Notification Endpoints

GET	/api/notifications	My notifications
PUT	/api/notifications/:id/read	Mark as read
PUT	/api/notifications/read-all	Mark all as read
PUT	/api/notifications/preferences	Update notification preferences
POST	/api/notifications/register-push	Register push token

7.9 User Endpoints

GET	/api/users/me	My profile
PUT	/api/users/me	Update profile
POST	/api/users/bank-accounts	Add bank account
DELETE	/api/users/bank-accounts/:id	Remove bank account
PUT	/api/users/bank-accounts/:id/primary	Set primary bank account
GET	/api/users/referral	My referral info
POST	/api/users/delete-account	Request account deletion

7.10 Admin Endpoints

GET	/api/admin/dashboard	Dashboard stats
-----	----------------------	-----------------

GET	/api/admin/investors	List all investors
GET	/api/admin/investors/:id	Investor detail
POST	/api/admin/investors/:id/freeze	Freeze account
POST	/api/admin/investors/:id/unfreeze	Unfreeze account
GET	/api/admin/kyc-queue	Pending KYC reviews
POST	/api/admin/kyc/:id/approve	Approve KYC
POST	/api/admin/kyc/:id/reject	Reject KYC
GET	/api/admin/developers	List developer partners
POST	/api/admin/developers/invite	Invite new developer
GET	/api/admin/transactions	All transactions
GET	/api/admin/financial-reports	Revenue, expenses, per-property
GET	/api/admin/audit-log	Audit log
GET	/api/admin/staff	List staff
POST	/api/admin/staff	Add staff member
PUT	/api/admin/staff/:id	Update staff role
DELETE	/api/admin/staff/:id	Deactivate staff
GET	/api/admin/settings	Platform settings
PUT	/api/admin/settings	Update settings
POST	/api/admin/announcements	Send announcement

7.11 Webhook Endpoints

POST	/api/webhooks/paystack	Handle all Paystack events: - charge.success (DVA payment received + auto-split to developer) - subaccount.credit (developer sub-account credited) - transfer.success (payout sent to investor) - transfer.failed (payout failed) - transfer.reversed (payout reversed)
------	------------------------	---

7.12 Middleware

Middleware	Purpose
auth	Verify JWT token, attach user to request
requireRole(roles[])	Check user role (investor, developer, admin)
requireAdminRole(roles[])	Check admin sub-role (ceo, coo, finance, etc.)
requireKYC(tier)	Check investor has minimum KYC tier
rateLimit	Rate limiting per IP/user
auditLog	Auto-log admin actions
validatePaystackWebhook	Verify Paystack webhook signature

8. PHASE MAPPING

What to build and when. Every feature tagged.

Phase 0: MVP (Month 1-3) — Launch with one property

Must have. Without these, you cannot launch.

Platform	Features
Investor App	Auth (signup, login, OTP), KYC Tier 1 (bank account), home dashboard, property browsing, property detail, investment flow (DVA payment), portfolio (holdings + activity), payouts (history + bank accounts), secondary market (list, buy, sell), profile (basic), notifications (transactional only)
Developer Portal	Auth, dashboard, property management, add property, rent/sale reporting, payout history, document management
Admin Portal	Dashboard, properties (list, approve/reject), investors (list, detail, freeze), KYC queue, payouts (calculate, approve, trigger, retry), P2P monitoring, developer management, staff management, system settings, audit log, broadcast alerts (push + in-app banner to all users)
Website	Landing page, for investors, for developers, FAQ, contact, developer application form, legal pages
Backend	All core API endpoints, Paystack DVA + Transfers integration, webhook handling, JWT auth, role middleware
Database	All core collections and indexes

Phase 1: Launch Features (Month 3-6) — Multiple properties, growing users

Nice to have. Improves experience significantly.

Platform	Features
Investor App	KYC Tier 2 (BVN), developer profiles + follow system, property updates tab, wishlist, share property, investment calculator, dark mode, biometric login, notification preferences, payout breakdown, portfolio performance chart, app announcements
Developer Portal	2FA, investor analytics (basic), follower growth, QR codes, property valuation updates, marketing toolkit (branded links)
Admin Portal	Property health flags, rent tracking + overdue alerts, communication log, financial reports (basic), KYC stats, notification templates, announcements + email blasts, staff activity log, feature flags, reconciliation
Website	About page

Phase 2: Growth Features (Month 6-18) — Engagement and intelligence

Growth accelerators. Builds retention and engagement.

Platform	Features
Investor App	KYC Tier 3 (NIN + selfie via Dojah), AI-recommended properties, engagement notifications, weekly portfolio summary, milestone celebrations, referral system, document vault, in-app support chat, marketplace price alerts, offline mode
Developer Portal	Social media graphics generator, embed widget, share analytics, investor demographics, conversion analytics
Admin Portal	Compliance dashboard, transaction monitoring, AML risk scores, STR filing, tax reports, scheduled announcements, investor segmentation
Website	Blog
AI/Smart	Recommendation engine, smart notification timing, fraud detection, investor segmentation

Phase 3: Scale (Month 18+) — Automation and advanced features

Platform	Features
All	Automated payout processing (remove manual approval for established properties), advanced analytics dashboards, API for third-party integrations, white-label options for large developer partners
Investor App	Investment goals/targets, auto-invest (recurring investment), property comparison tool
Admin	Automated compliance reporting, developer scoring system, predictive analytics

9. THIRD-PARTY INTEGRATIONS

Service	Provider	Purpose	Phase	Cost
Payment collection	Paystack DVA	Investor payments via bank transfer	MVP	1% capped N300/tx
Payment split	Paystack Sub-accounts	Auto-split: 97% to developer, 3% to PropVest	MVP	Included in DVA fee
Payouts	Paystack Transfers	Send money to investor bank accounts	MVP	N10-50/tx
KYC Tier 1	Paystack Resolve Account	Verify bank account name	MVP	Free
KYC Tier 2	Paystack BVN Resolve	Verify BVN	Phase 1	N10/call
KYC Tier 3	Dojah	NIN + facial verification	Phase 2	~N80-150/call
Email	Resend	Transactional + marketing emails	MVP	Free tier: 3K/month

Service	Provider	Purpose	Phase	Cost
SMS / OTP	Termii	Phone verification, payout alerts	MVP	~N4/SMS
Push notifications	Expo Push	Mobile push notifications	MVP	Free
Image storage	Cloudinary	Property photos, documents	MVP	Free tier: 25K transforms
Analytics	PostHog	User behavior tracking (for AI recs)	Phase 1	Free self-hosted
Monitoring	Sentry	Error tracking and performance	MVP	Free tier

10. AI / SMART FEATURES

10.1 Property Recommendation Engine (Phase 2)

Data signals used:

- Investment history (which property types, locations, developers, price ranges)
- Browsing behavior (properties viewed, time spent, search queries)
- Followed developers
- Wishlist items
- Demographic similarity (investors with similar profiles)

Algorithm (simple to start):

1. Content-based: recommend properties similar to ones you've invested in (same type, location, developer, price range)
2. Collaborative: "investors who bought X also bought Y"
3. Popularity: trending properties with high investment velocity
4. New from followed: always surface new listings from followed developers

Implementation: Start with simple rule-based matching (Phase 2). Move to ML model if needed (Phase 3+). PostHog event tracking provides the behavioral data.

10.2 Smart Notification Timing (Phase 2)

Track when each user typically opens the app → send engagement pushes during that window.
Default: 9am-12pm for most Nigerian users.

10.3 Fraud Detection (Phase 2)

Flags:

- Same bank account on multiple investor accounts
- Rapid buy/sell cycles on marketplace (potential wash trading)
- Investment amount suddenly far above historical pattern
- Multiple failed payment attempts
- Account created and immediately invests maximum amount

Action: Auto-flag for compliance review. Don't auto-block — humans decide.

APPENDIX: SCREEN COUNT SUMMARY

Platform	Screens/Pages	Phase
Investor App	~25 screens	MVP: 15, Phase 1: 7, Phase 2: 3
Developer Portal	~12 pages	MVP: 8, Phase 1: 3, Phase 2: 1
Admin Portal	~18 pages	MVP: 12, Phase 1: 4, Phase 2: 2
Product Website	~10 pages	MVP: 7, Phase 1: 1, Phase 2: 2
Total	~65 screens/pages	

APPENDIX: MVP CHECKLIST (What to build FIRST)

Build these and ONLY these before launching with the first property.

- [] Investor signup + OTP + login
- [] KYC Tier 1 (bank account verification)
- [] Property browsing (list + detail)
- [] Investment flow (select units → DVA payment → confirmation)
- [] Paystack webhook handler (auto-allocate units on payment)
- [] Portfolio (my holdings + activity)
- [] Payouts page (history + bank account management)
- [] Secondary market (list, browse, buy, sell)
- [] Basic notifications (transactional: payment confirmed, payout sent)
- [] Profile + settings (basic)
- [] Developer portal: login, dashboard, add property, manage property, report rent/sale
- [] Admin portal: dashboard, properties (approve/reject), investors (list, freeze), KYC queue, payouts (calculate, approve, trigger), P2P monitoring, developer management, staff RBAC, settings, audit log
- [] Product website: landing, for investors, for developers, FAQ, contact, legal pages
- [] Paystack DVA integration (collections)
- [] Paystack Transfers integration (payouts)
- [] Paystack Resolve Account integration (KYC)
- [] Expo Push Notifications (basic)
- [] Resend email integration (transactional)
- [] Termii SMS integration (OTP)

Estimated screens for MVP: ~42 screens/pages across all platforms.

This PRD should be reviewed and updated as the product evolves. Version: 1.0 — April 2026